Proxmox VE 3-Node Architecture & Health Verification

Cluster ID: **cisai-pve** • Lead Engineer: **Nishu Saini** • Organization: **Innspark / CISAI**

100% LIVE VERIFIED HEALTHY

CLUSTER QUORUM	CEPH SHARED STORAGE	ZFS RAIDZ2 STORAGE	HIGH AVAILABILITY
3 / 3 Quorate	**12 / 12 OSDs Up**	**ONLINE (0 Errors)**	**CRM Watchdog Armed**
Votes: 3 | Quorum Req: 2	218 TiB Raw | 33 Clean PGs	3x Pools (~90 TiB per host)	3 Managed VMs Running

1. Live Architecture Topology Diagram

LAN 1: Corporate Management & Corosync Heartbeat Subnet (10.31.6.0/22 on vmbr0) • Gateway: 10.31.4.1 — Latency: < 0.3 ms

prox22 — HA Master

Mgmt IP: `10.31.6.172` (vmbr0)
10G Backplane: `192.168.10.22` (nic3)
Role: Active HA Master • Active Ceph MGR
Ceph: OSD 0, 3, 6, 9 (All UP) • MON 2
ZFS: `zfs-ha/pve` (Replication Source)

VM 9002 (Running)

prox23 — Member

Mgmt IP: `10.31.6.173` (vmbr0)
10G Backplane: `192.168.10.23` (nic3)
Role: Quorum Member • Standby MGR
Ceph: OSD 1, 4, 7, 10 (All UP) • MON 3
ZFS: `zfs-ha/pve` (3.5s Sync Target)

VM 9003 (Running) VM 101 (Running)

prox24 — MON Leader

Mgmt IP: `10.31.6.174` (vmbr0)
10G Backplane: `192.168.10.24` (nic3)
Role: Active Ceph MON Leader
Ceph: OSD 2, 5, 8, 11 (All UP) • MON 1
ZFS: `zfs-ha/pve` (3.3s Sync Target)

VM 9001 (Running) VM 100 (Running)

LAN 2: Dedicated 10G SFP+ Storage & RAM Migration Backplane (192.168.10.0/24 on nic3) • DAC — Throughput: ~9.4 Gbps • Latency: ~0.07 ms

2. Live Cluster Physical Nodes & Network Mapping

Node Name	Node ID	Management IP	10G Interconnect IP	Corosync Link	Ceph OSDs Hosted	Cluster Role	Live Status
prox22	`0x0002`	`10.31.6.172`	`192.168.10.22`	Link 0 (1 Vote)	osd.0, osd.3, osd.6, osd.9	HA Master • Active Ceph MGR	**ONLINE / QUORATE**
prox23	`0x0003`	`10.31.6.173`	`192.168.10.23`	Link 0 (1 Vote)	osd.1, osd.4, osd.7, osd.10	Member • Standby MGR	**ONLINE / QUORATE**
prox24	`0x0001`	`10.31.6.174`	`192.168.10.24`	Link 0 (1 Vote)	osd.2, osd.5, osd.8, osd.11	Ceph MON Leader	**ONLINE / QUORATE**

3. Dual-Tier Storage Subsystems

Storage Tier	Storage ID	Technology	Total Capacity	Redundancy Profile	Operational Purpose	Health
Tier 1 (Shared)	`ceph`	Ceph RBD Distributed Engine	218 TiB Raw (69 TiB Avail)	3x Replicated (size=3, min=2)	Zero-copy live migration, HA zero-RPO storage	**HEALTH_OK (33 Clean PGs)**
Tier 2 (Local)	`zfs-ha`	OpenZFS 7-Disk RAIDZ2 Pool	~90 TiB Usable / Host	Dual Parity (2-drive failure survival)	Extreme OLTP IOPS (47K IOPS), ARC memory caching	**ONLINE (0 Errors)**
Backup / ISO	`NFS-Storage`	NFS on 10.31.1.161	2.8 TiB	External NAS Appliance	Template ISOs, rootfs archives, VM backups	**ACTIVE**

Active Workload & High Availability Verification

Live Proxmox HA Manager • Virtual Machine Distribution • Network Isolation Rules

CRM WATCHDOG ARMED

4. Live Virtual Machine Workload Distribution

VM ID	VM Name	Host Node	Storage Backend	Compute Sizing	HA Policy	Replication / Live Mobility
9001	`ha-test-ceph-ubuntu2204`	**prox24**	`ceph:vm-9001-disk-1`	2 vCPU / 4096 MB	CRM (started)	**Instant Zero-Copy 10G Live Migration**
9002	`zfs-test-prox24-ubuntu2204`	**prox22**	`zfs-ha:vm-9002-disk-0`	2 vCPU / 4096 MB	CRM (started)	**Auto-Replicated to prox23 & prox24 (3.4s sync)**
9003	`zfs-benchmark-prod-vm`	**prox23**	`zfs-ha:vm-9003-disk-1`	2 vCPU / 4096 MB	CRM (started)	Production IOPS Benchmark & Snapshot Target
100	`101-sajanlals-ubuntu24`	**prox24**	`local-lvm:vm-100-disk-0`	4 vCPU / 5120 MB	Manual	Local LVM (Static IP: 10.31.4.235)
101	`102-sajanlals-ubuntu24`	**prox23**	`local-lvm:vm-101-disk-0`	3 vCPU / 5120 MB	Manual	Local LVM (Static IP: 10.73.34.43)

5. High Availability & Fencing Configuration

Fencing & CRM Architecture: The cluster operates with `corosync_votequorum` (3 votes, 2 required for quorum). The Master CRM runs on `prox22` with hardware/softdog watchdog fencing armed on all nodes. In case of unexpected host disconnection, the active CRM watchdog guarantees split-brain protection and restarts protected VMs on surviving nodes.

HA Subsystem	Daemon Process	Active Node	Watchdog State	Failover / Synchrony Performance
Cluster Resource Manager	`pve-ha-crm`	**prox22** (Master)	Armed (Active Watchdog)	~15 seconds lock takeover on leader death
Local Resource Manager	`pve-ha-lrm`	prox22, prox23, prox24	Armed (Active Watchdog)	Immediate local process supervision
Automated ZFS Replication	`pvesr`	Managed Proxmox Schedule	Active (Every 30 mins)	3.38s – 3.52s delta transfer time (Job 9002-0/1)

6. 10G Network Isolation & Live Migration Policy

```
keyboard: en-us • ha: shutdown_policy=migrate • migration: type=secure,network=192.168.10.0/24
```

Operational Metric	Observed Performance	Architectural Impact
Peak Live Migration Throughput	**828.6 MiB/s (~6.95 Gbps)**	RAM live migration fully saturates the 10G link without dropping packets.
Guest Cutover Downtime	**23 milliseconds**	Imperceptible guest freeze during memory state handoff; zero dropped pings.
Corosync Heartbeat Protection	**0% Load on 1G Management LAN**	Heavy RAM replication traffic is strictly barred from `vmbr0` (10.31.6.0/22).

7. Verification Sign-Off & Engineering Approval

Review Item	Audited By	Date	Verdict
Corosync Quorum & Node Health	Nishu Saini (Innspark)	2026-09-19	**PASS • 100% OPERATIONAL**
Ceph 12-OSD Storage Health	Nishu Saini (Innspark)	2026-09-19	**PASS • HEALTH_OK**
ZFS RAIDZ2 & Cross-Node Sync	Nishu Saini (Innspark)	2026-09-19	**PASS • SYNCHRONIZED**
High Availability CRM Fencing	Nishu Saini (Innspark)	2026-09-19	**PASS • ARMED**